Huawei Access Node QoS/Congestion Event Troubleshooting Procedure for Whitefield POP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3143

Description

The QoS/Congestion Event failure domain, characterized by microbursts, interface tail drops, and QoS queue starvation, is a critical issue affecting the Access role on Huawei equipment at the Whitefield POP. This occurs due to the complex interplay of packet scheduling, queuing, and buffering mechanisms within the network device. Specifically, the Huawei's Versatile Routing Platform (VRP) utilizes a combination of Priority Queuing (PQ), Class-Based Queuing (CBQ), and Weighted Round Robin (WRR) algorithms to manage traffic. However, under certain conditions such as sudden changes in traffic patterns, misconfigured QoS policies, or hardware resource constraints, these mechanisms can lead to congestion and subsequent packet drops. The Huawei CLI dialect provides a robust set of commands to diagnose and resolve these issues, leveraging the device's control and data plane components. Understanding the underlying protocol and hardware mechanics, such as the role of the Network Processor Units (NPUs) in traffic management, is crucial for effective troubleshooting and resolution of QoS/Congestion Events on Huawei Access nodes.

Pre-requisite Checks

- Dispatch remote hands to Whitefield POP for physical inspection and potential hardware replacement
- Verify redundant link status to ensure failover mechanisms are operational
- Check for any recent software or hardware changes that may have introduced the issue
- Ensure all relevant network management and monitoring tools are accessible and functional

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Hardware resource constraints (e.g., insufficient buffer space)	Medium
Sudden changes in traffic patterns	Low

Diagnostic Commands & Expected Outputs

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Command: display qos interface, Expected Output: QoS interface configuration and statistics, Description: This command is used to display the current QoS configuration and statistics for the specified interface, allowing for the identification of potential QoS policy misconfigurations or interface-specific issues.

Command: display cpu-usage, Expected Output: CPU usage statistics, Description: This command provides insights into the device's CPU
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utilization, helping to identify if the congestion event is related to high CPU usage, possibly due to software issues or excessive control plane traffic.

Command: display buffer threshold, Expected Output: Buffer threshold configuration and status, Description: The display buffer threshold command is crucial for understanding the buffer management on the device, including threshold settings and current buffer utilization, which can indicate if buffer exhaustion is contributing to the congestion.
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Resolution Steps

Step 1: Step: Adjust QoS Policies, Command: qos policy class priority , Description: Adjusting the QoS policies to ensure proper prioritization and allocation of bandwidth can resolve issues related to misconfigured QoS. This involves modifying the existing policy to better match the current traffic requirements and patterns.

Step 2: Step: Increase Buffer Space, Command: buffer threshold , Description: If diagnostic commands indicate buffer space as a limiting factor, increasing the buffer threshold can provide temporary relief from congestion. However, this should be done with caution and thorough understanding of the potential impact on other traffic classes.

Step 3: Step: Optimize Traffic Management, Command: traffic-manage shaping , Description: Optimizing traffic management involves configuring shaping and policing to better regulate the flow of traffic into the network, preventing sudden spikes that can cause congestion. This step requires careful analysis of traffic patterns and capacity planning.

Post-Check Verification

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Command: display qos interface, Expected Output: Updated QoS interface configuration and statistics reflecting changes, Description: After making adjustments to QoS policies or buffer thresholds, re-running diagnostic commands to verify that the changes have taken effect and are improving the congestion situation is essential.

Command: display cpu-usage, Expected Output: Stable CPU usage statistics, Description: Verifying that CPU usage has stabilized or decreased after implementing fixes helps confirm that the resolution steps have not introduced new issues and that the device is operating within expected parameters.
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Roll-back Procedures

- Step: Revert QoS Policy Changes, Command: qos policy class priority , Description: If the applied fixes do not resolve the issue or introduce new problems, reverting the QoS policy changes to their original state is necessary to avoid further complicating the situation.

- Step: Restore Original Buffer Threshold, Command: buffer threshold , Description: Similarly, if increasing the buffer threshold did not alleviate congestion or caused unintended side effects, restoring the original buffer threshold setting is required to maintain consistency with the pre-existing configuration.